Jongmin Choi

M.S. Student in Electrical Engineering · Multimodal AI (MMAI) Lab, KAIST

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RESEARCH INTERESTS

Biological intelligence provides the clearest evidence that learning can be both general and efficient, and I believe that building its principles into model architectures matters as much as scaling them up. From this perspective, I have studied multimodal learning that bridges the gap between audio and text while preserving the structure unique to each modality, as well as spiking neurons with temporal dynamics. Currently, guided by the idea that good reasoning begins with good input selection, I am exploring architectures that dynamically extract query-relevant information from diverse modalities and make it readily usable by language models. In parallel, I investigate the mathematical principles underlying deep network optimization.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) — *Daejeon, South Korea*

Ph.D. in Electrical Engineering (Incoming) — *Starting Mar 2027*
Advisor: **Prof. Joon Son Chung** · Multimodal AI (MMAI) Lab

M.S. in Electrical Engineering — *Mar 2025 – Feb 2027 (expected)*
Advisor: **Prof. Joon Son Chung** · Multimodal AI (MMAI) Lab · GPA **4.06 / 4.30**

B.S. in Mathematical Sciences and Electrical Engineering (Double Major) — *Mar 2018 – Feb 2025*
GPA **3.65 / 4.30**

PUBLICATIONS

** Equal contribution*

NeurIPS 2026 ★ SPOTLIGHT
FiTS: Interpretable Spiking Neurons via Frequency Selectivity and Temporal Shaping
Jongmin Choi, Joon Son Chung
Conference on Neural Information Processing Systems · **0.95%** (292 / 30,709)

ICASSP 2026 ★ ORAL
LAMB: LLM-based Audio Captioning with Modality Gap Bridging via Cauchy–Schwarz Divergence
Hyeongkeun Lee*, **Jongmin Choi***, KiHyun Nam, Joon Son Chung
IEEE International Conference on Acoustics, Speech, and Signal Processing

ICASSP 2026 POSTER
Diffusion-Link: Diffusion Probabilistic Model for Bridging the Audio–Text Modality Gap
KiHyun Nam*, **Jongmin Choi***, Hyeongkeun Lee, Jungwoo Heo, Joon Son Chung
IEEE International Conference on Acoustics, Speech, and Signal Processing

EMNLP 2026 *Findings* POSTER
Fork-Merge Decoding: Enhancing Multimodal Understanding in Audio-Visual Large Language Models
Chaeyoung Jung*, Youngjoon Jang*, **Jongmin Choi**, Joon Son Chung
Findings of Empirical Methods in Natural Language Processing

arXiv 2026 PREPRINT
OmniKVQuant: KV Cache Quantization for Omni-LLMs
Suho Yoo*, Hyunjong Ok*, **Jongmin Choi**, Jihoo Jung, Joon Son Chung
Under review at IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2027)

TEACHING EXPERIENCE

EE.70038 — Speech Recognition Systems — *KAIST* — *Spring 2026*
Teaching Assistant · Instructor: **Prof. Joon Son Chung**

EE.40034 — Deep Learning for Visual Understanding — *KAIST* — *Fall 2025*
Teaching Assistant · Instructor: **Prof. Joon Son Chung**

SKILLS

Research & ML: Python, PyTorch · **iOS Development:** Swift, SwiftUI, SwiftData · **Languages:** Korean (native), English (professional working proficiency)